

2013-DSE
PHY
PAPER 1B

B

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2013

PHYSICS PAPER 1

SECTION B: Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

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Please stick the barcode label here.

Candidate Number

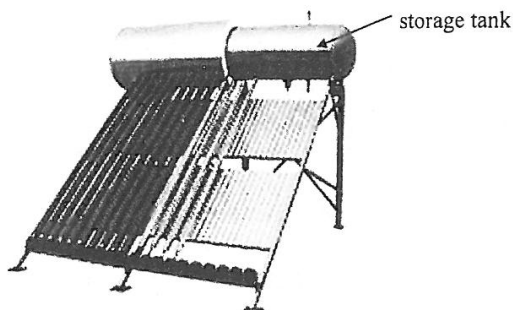
Question No.	Marks
1	6
2	6
3	8
4	7
5	5
6	9
7	10
8	9
9	6
10	11
11	7



Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

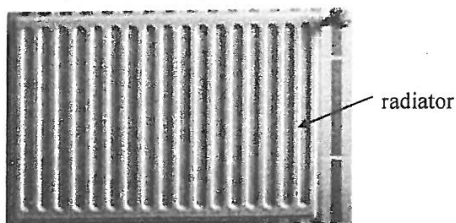
1.

Figure 1.1



A solar water heater shown in Figure 1.1 is installed on the rooftop of a house. During the day, the heater heats up 1.5 m^3 of water to 80°C . At night, the hot water in the storage tank is circulated to the radiators (see Figure 1.2) in different rooms of the house to keep the rooms warm.

Figure 1.2



Given: density of water = 1000 kg m^{-3}
specific heat capacity of water = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$

- (a) Given that 15% of the energy is lost during the transfer of water, how much heat can be released from the system to the rooms when the water temperature drops to 60°C ? (3 marks)

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2014-DSE
PHY
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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2014

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Candidate Number

Question No.	Marks
1	7
2	6
3	12
4	9
5	7
6	8
7	7
8	12
9	8
10	8

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1.

Figure 1.1

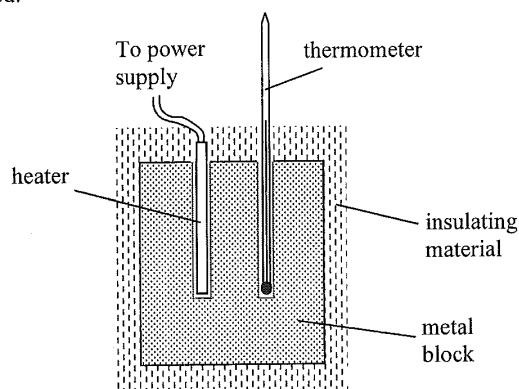
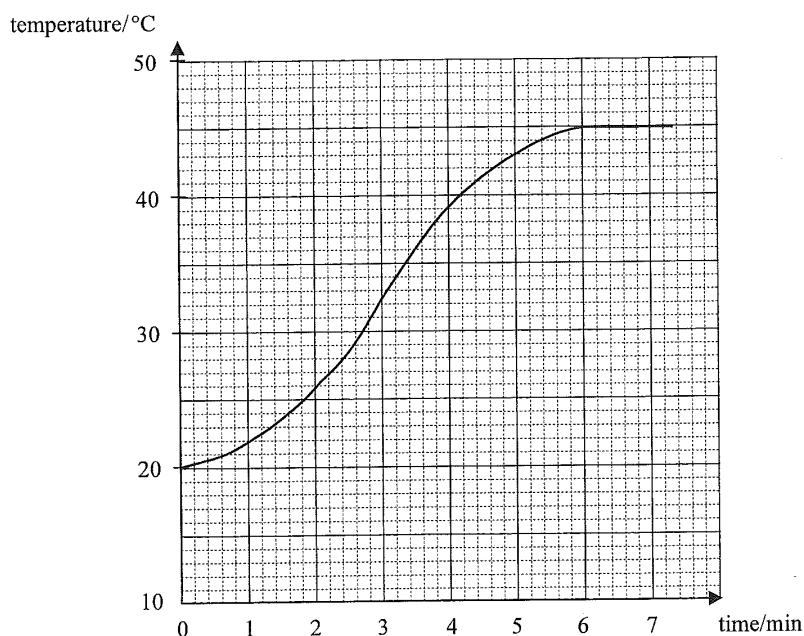


Figure 1.1 shows an experimental set-up to find the specific heat capacity of a metal. The metal block is wrapped by insulating material. A heater is connected to a power supply. It is switched on when the temperature of the metal block is 20°C and then switched off when the temperature reaches 43°C . The graph below shows the variation of the temperature of the metal block with time.



- (a) Use the graph to find the duration that the heater is switched on. (1 mark)

- (b) After the heater is switched off, the temperature of the metal block continues to rise for a while. Explain why. (1 mark)

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1	8
2	7
3	9
4	11
5	11
6	10
7	9
8	11
9	8



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2016-DSE-PHY 1B-2

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2018-DSE
PHY
PAPER 1B

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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2018

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PHYSICS PAPER 1

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Question No.	Marks
1	5
2	6
3	5
4	11
5	9
6	10
7	9
8	9
9	12
10	8



2020-DSE
PHY

PAPER 1B

B

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2020

PHYSICS PAPER 1

SECTION B : Question-Answer Book B

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Candidate Number

Question No.	Marks
1	8
2	13
3	6
4	10
5	8
6	6
7	9
8	11
9	8
10	5

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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. In a restaurant, 'wontons in soup' is prepared by putting 5 pieces of cooked wonton at 4°C into a bowl with 0.60 kg of soup at temperature 96°C .

Given: average mass of each piece of wonton = 0.02 kg
 specific heat capacity of wonton = $3300\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$
 specific heat capacity of soup = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (a) Find the final temperature of the mixture. Assume that the heat capacity of the bowl and the heat loss to the surroundings are negligible. (2 marks)

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- (b) The soup in (a) is taken from a metallic container of heat capacity $2000\text{ J }^{\circ}\text{C}^{-1}$ containing 16 kg of soup maintained at 96°C by an immersion heater.

- (i) Why does that energy have to be supplied by the heater to keep the soup at 96°C ? (1 mark)

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2023-DSE
PHY
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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2023

PHYSICS PAPER 1

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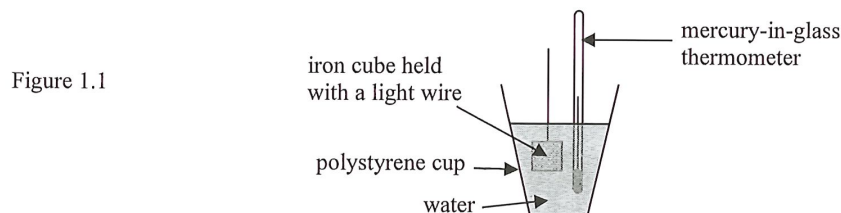
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Question No.	Marks
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3	12
4	10
5	8
6	9
7	8
8	11
9	7



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1. An iron cube of mass 200 g is heated to a few hundred degree Celsius (say at $T^{\circ}\text{C}$). It is then quickly transferred into a polystyrene cup containing 600 g of water. The cube is totally immersed in the water as shown in Figure 1.1. The temperature of the heated iron cube can be estimated in this experiment.



Assume that no heat is lost to the surroundings and neglect the heat capacity of the polystyrene cup.

- (a) The temperature of water recorded by the thermometer rises from 25°C to 33°C . Estimate T .
Given: The specific heat capacities of iron and water are $450 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ respectively. (3 marks)

- (b) What would happen to the water when this iron cube of high temperature just touches the water surface? (1 mark)

- (c) Explain why the actual temperature of the iron cube should be higher than that estimated in (a). (2 marks)

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2025-DSE
PHY
PAPER 1B

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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2025

PHYSICS PAPER 1

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Short Questions Question No.	Marks
1	5
2	6
3	6
4	4
5	4
6	8
7	6
8	6
9	6
Long Questions Question No.	Marks
10	12
11	12
12	9

